

Week 4: Machine Learning Algorithms & Model Evaluation

Domain: Artificial Intelligence | Batch: 6-Week AI Internship

Duration: 1 Week (7 Days) | Total Marks: 100

Task 1: Machine Learning Algorithms

Four algorithms are explained below: two classification algorithms used in Task 2 (Logistic Regression, Decision Tree Classifier) and Linear Regression used in Task 3, plus K-Nearest Neighbors as a fourth, widely-used comparison algorithm.

1. Logistic Regression

What it is: A supervised classification algorithm that predicts the probability of a binary (or multi-class) outcome, despite the word 'Regression' in its name.

How it works: It computes a weighted sum of the input features, then passes that sum through a sigmoid function to squeeze the result into a probability between 0 and 1. If the probability is above a threshold (typically 0.5), the model predicts the positive class; otherwise the negative class. Weights are learned by minimizing a loss function (log-loss) during training.

Where it is used: Any binary classification task with a roughly linear relationship between features and outcome: medical diagnosis, credit approval, spam detection, pass/fail prediction.

Real-world application: Used in this assignment (Task 2) to classify breast tumor samples as malignant or benign based on 30 cell-nucleus measurements, achieving 98.25% test accuracy.

Advantages: Simple, fast to train, highly interpretable (coefficients show each feature's influence), works well even with modest amounts of data, and outputs calibrated probabilities rather than just a hard label.

Limitations: Assumes a roughly linear decision boundary, so it can underperform on data with complex, non-linear relationships between features and outcome.

2. Decision Tree Classifier

What it is: A supervised classification algorithm that makes predictions by learning a sequence of if-else questions about the input features, structured as a tree.

How it works: Starting from a root node containing all training data, the algorithm repeatedly splits the data on the feature and threshold that best separates the classes (commonly measured using Gini impurity or information gain), continuing until a stopping condition is met (e.g. a maximum depth). To predict a new sample, it is passed down the tree, answering each question, until it reaches a leaf node that gives the final predicted class.

Where it is used: Situations where an interpretable, rule-based decision process is valuable: loan approval, medical triage, customer segmentation, and as a building block for ensemble methods like Random Forests.

Real-world application: Used in this assignment (Task 2 and the Task 5 mini project) to classify breast cancer samples and predict student Pass/Fail outcomes as a comparison model against Logistic Regression.

Advantages: Easy to visualize and explain (a literal flowchart of decisions), handles non-linear relationships naturally, requires little data preprocessing (no feature scaling needed).

Limitations: Prone to overfitting on training data if the tree is allowed to grow too deep (mitigated here with `max_depth` limits); small changes in data can produce a very different tree, making individual trees less stable than ensemble methods.

3. Linear Regression

What it is: A supervised regression algorithm that predicts a continuous numeric value by fitting a straight-line (linear) relationship between the input features and the target.

How it works: The model learns a coefficient (weight) for each feature and an intercept, such that the weighted sum of features plus the intercept best approximates the target value across the training data. It is typically trained by minimizing the mean squared error between predicted and actual values (Ordinary Least Squares).

Where it is used: Any problem where the goal is to predict a continuous number from one or more input variables: price prediction, sales forecasting, risk scoring, trend estimation.

Real-world application: Used in this assignment (Task 3) to predict house prices from area, bedrooms, bathrooms, age, and distance to the city, achieving an R2 score of about 0.92 on the test set.

Advantages: Simple, fast, highly interpretable (each coefficient shows how much the target changes per unit change in a feature), and works well when the true relationship is approximately linear.

Limitations: Cannot capture non-linear relationships well, sensitive to outliers, and assumes features are not too strongly correlated with each other (multicollinearity can distort coefficient estimates).

4. K-Nearest Neighbors (KNN)

What it is: A supervised algorithm usable for both classification and regression, which makes predictions based on the 'k' most similar examples in the training data.

How it works: For a new data point, KNN calculates the distance (commonly Euclidean distance) to every point in the training set, selects the k closest neighbors, and predicts based on them -- the majority class for classification, or the average value for regression. There is no explicit 'training' phase beyond storing the data; all computation happens at prediction time.

Where it is used: Recommendation systems, image recognition, anomaly detection, and any problem where similar inputs are expected to have similar outputs.

Real-world application: A commonly used baseline for comparison against Logistic Regression / Decision Trees on classification tasks such as this assignment's breast cancer and student performance problems.

Advantages: Simple and intuitive, makes no assumptions about the underlying data distribution, naturally handles multi-class problems.

Limitations: Computationally expensive at prediction time on large datasets (must compare against every training point), sensitive to feature scaling and irrelevant features, and performance degrades in high-dimensional spaces (the 'curse of dimensionality').

Task 2: Build Classification Models

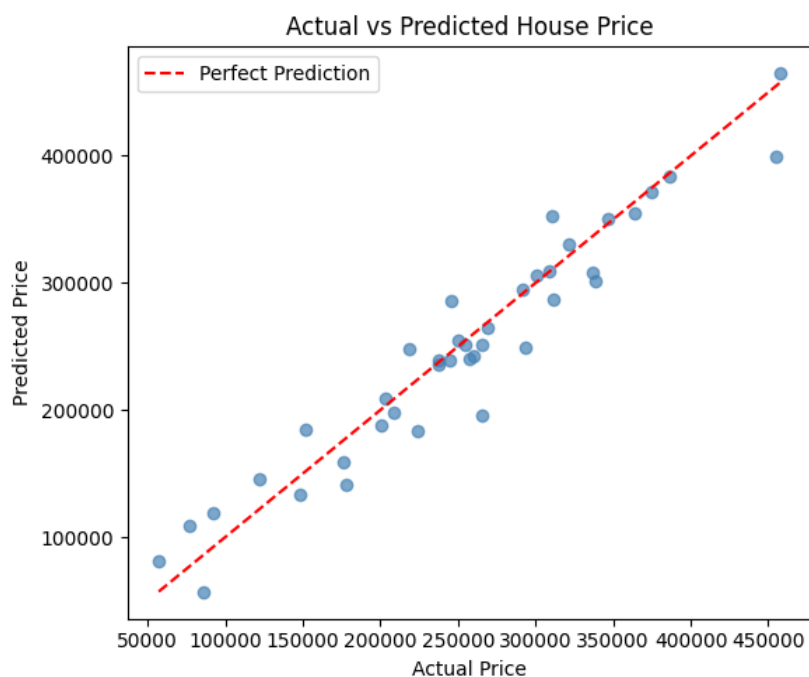
Dataset: Breast Cancer Wisconsin Diagnostic Dataset (via `sklearn.datasets.load_breast_cancer`) -- 569 samples, 30 numeric features describing cell nuclei from digitized images of breast tissue, target: malignant (0) or benign (1).

The dataset was loaded, explored (no missing values or duplicates were found), and split into features (X, 30 columns) and target (y). Data was split 80/20 into training and testing sets (stratified to preserve the class balance), and features were standardized for Logistic Regression. Two models were trained: Logistic Regression and a depth-limited Decision Tree Classifier. Both were used to make predictions on the held-out test set.

Task 3: Build a Regression Model

Dataset: House Price Dataset (synthetically generated for this assignment, 200 samples) -- features: Area (sq ft), Bedrooms, Bathrooms, Age (years), Distance to City (km); target: Price.

A small number of missing values were filled with column means. The data was split 80/20 into training and testing sets, and a Linear Regression model was trained to predict price from the five features. Predictions were compared against actual prices, and a scatter plot of Actual vs Predicted price was created to visualize model performance.



Actual vs Predicted house price. Points near the dashed red line indicate accurate predictions.

Task 4: Model Evaluation & Comparison

Classification Models (Breast Cancer Dataset)

Model	Accuracy	Precision	Recall
Logistic Regression	98.25%	98.61%	98.61%
Decision Tree Classifier	92.11%	95.65%	91.67%

Regression Model (House Price Dataset)

Model	MAE	RMSE	R2 Score
Linear Regression	19,825	25,922	0.924

Which model performed better and why?

For the breast cancer classification task, **Logistic Regression outperformed the Decision Tree** (98.25% vs 92.11% accuracy). This is because the classes in this dataset are fairly linearly separable in feature space -- most tumor measurements (size, texture, smoothness) scale fairly smoothly with malignancy -- which plays to Logistic Regression's strengths. The Decision Tree, while still strong, is more prone to overfitting fine-grained noise in the training split even with a limited depth.

Which model would be selected: Logistic Regression is the preferred choice here -- it is more accurate, simpler, faster, and its coefficients are directly interpretable, which matters in a medical diagnostic context where a doctor may need to understand why a prediction was made.

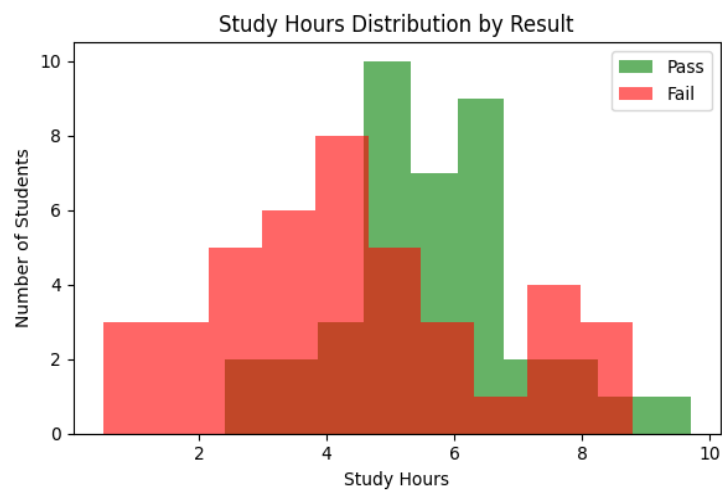
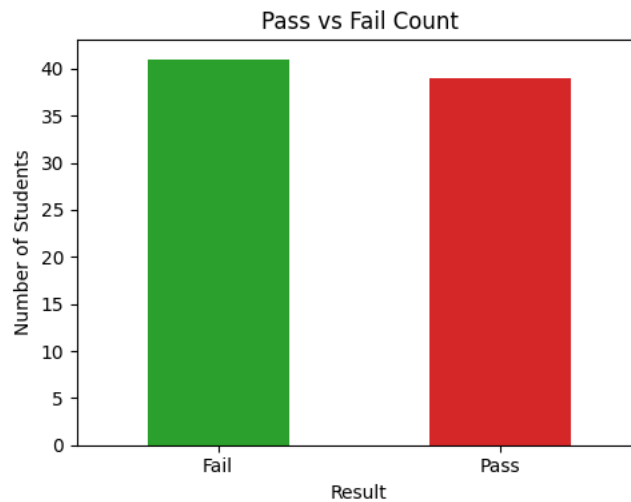
Limitations of the selected model: Logistic Regression assumes an approximately linear decision boundary, so it could underperform if the true relationship between cell features and malignancy were more complex/non-linear. Results also come from a single train-test split, so cross-validation would give a more robust accuracy estimate. For the regression model, an R2 of 0.924 indicates the model explains most (but not all) of the variance in house prices -- some price variation from unmodeled factors (e.g. neighborhood quality, renovation status) is not captured by the five features used.

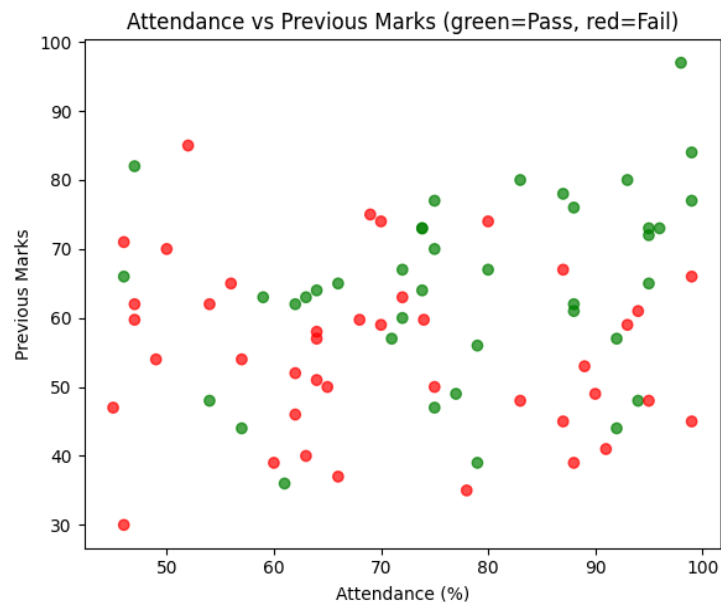
Task 5: Mini Project - ML Model Comparison

Project: Student Performance Prediction

Problem statement: Predict whether a student will Pass or Fail based on study hours, attendance, previous marks, and assignment score -- useful for early identification of at-risk students.

Dataset used: 80 students (synthetically generated for this assignment, continued from Week 3), with a small number of missing values filled using column means during cleaning.





Models selected: Logistic Regression and a depth-limited Decision Tree Classifier -- the same two algorithms compared in Task 2, applied here to a second, independent dataset and problem.

Model	Accuracy	Precision	Recall
Logistic Regression	56.25%	50.00%	57.14%
Decision Tree Classifier	75.00%	100.00%	42.86%

Results: The Decision Tree achieved higher accuracy (75.00% vs 56.25%) and perfect precision (100%) on this small test set, meaning every student it predicted would Pass, actually passed -- but its lower recall (42.86%) means it missed some students who did in fact pass, predicting Fail for them instead. Logistic Regression was more balanced but less accurate overall.

Best-performing model: The Decision Tree Classifier, based on test accuracy and precision, though its low recall is a meaningful caveat -- for an early-warning system, missing genuine pass cases is less costly than missing genuine fail cases, so this trade-off should be considered alongside raw accuracy.

Limitations: With only 80 total students (16 in the test set), both accuracy figures carry substantial uncertainty, and the dataset includes intentional randomness in how Pass/Fail was generated, capping achievable accuracy for any model.

Future improvements: A larger, real dataset would improve reliability; cross-validation would give a more robust accuracy estimate than a single train-test split; and ensemble methods (Random Forest, Gradient Boosting) could be explored for potentially stronger, more balanced performance than either individual model tested here.

Deliverables Included

- This PDF report (Task 1 algorithm explanations + Tasks 2-5 summary, results, and comparisons)
- Jupyter Notebook (Week4_ML_Project.ipynb) - Tasks 2-5, fully executed with outputs and charts
- Datasets: breast_cancer_dataset.csv, house_price_dataset.csv, student_ml_dataset.csv